

Which Pass Rate Answers Which Question?

Unit 2 · Topic 2.2 · THE PROBLEM

Suppose a school records exam outcomes for 300 students. Students chose for themselves whether to use the tutoring center.

Mara: *The pass rate is 75%.*

Noah: *The pass rate is 88%.*

Elena: *Tutoring raises the pass rate by 13 percentage points because $88\% - 75\% = 13$ percentage points.*

Exam outcomes for 300 students

Center?	Passed	Did not pass	Total
Used	66	9	75
Did not	159	66	225
Total	225	75	300

FIRST TAKE — one sentence before you work the parts

Can 75% and 88% both be correct pass rates? State what group you think each percentage describes.

- DOK 1** (a) Label 75 percent and 88 percent as marginal or conditional. Name the group for each.
- DOK 2** (b) Verify the 75 percent and 88 percent rates. The nonuser pass rate is about 70.7 percent; find the user-minus-nonuser gap.
- DOK 3** (c) Can Mara and Noah both be right? Explain their denominators. Use the two group rates to correct Elena's 13-point comparison, and explain why her word raises goes beyond these data. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.